INCH-POUND

MIL-I-46224E (AR)
4 FEBRUARY 1989
SUPERSEDING
MIL-I-46224D
18 March 1975

MILITARY SPECIFICATION

IGNITER, ELECTRIC, PRIMER, M63, PARTS FOR, AND LOADING ASSEMBLING AND PACKING

This Specification is approved for use within the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the parts for, and the loading, assembling and packing for the Igniter, Electric, Primer, M63.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Packaging, Methods of
MIL-P-223	- Powder, Black
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1390

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1234	- Pyrotechnics, Sampling, Inspection and Testing

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8837897	-	Box, Set-up, Packing, Ammunition For Igniter, Electric, Primer M63
8837898	-	Box, Packing, Ammunition for Igniter, Electric, Primer, M63
8839497	-	Igniter, Electric, Primer, M63,
8839502	-	Closing Plug Loading Assembly
8839503	-	Plug, Closing
8839504	-	Body
8839505	-	Ignition Element Assembly
8839509	-	Retainer
9235760	-	Igniter, Electric, M63 Metal Parts

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein. (See 4.3).

3.2 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.3 Metal parts. Component parts shall comply with all requirements specified on Dwg. 9235760 and in applicable specifications.

3.4 Igniters. Igniters shall comply with all requirements specified on Dwg. 8839497 and in applicable specifications.

3.5 Moisture content

3.5.1 Ignition element charge. The moisture content of the ignition element charge after drying shall not exceed 0.02 percent.

3.5.2 Black powder. The moisture content of the black powder at the time of loading the closing plug shall not exceed 0.50 percent.

3.6 Electrical resistance. Electrical resistance between the ignition element and the body shall be not less than 0.80 ohms nor more than 1.80 ohms.

3.7 Functioning. The igniter shall function when a current of 1.25 amperes, maximum, is applied.

3.8 Non-functioning. The igniter shall not function when a current limited to 190 milliamperes, minimum, is applied.

3.9 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification.

4.2.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.3 Classification of inspection. The following types of inspections shall be conducted on this item.

a. First Article Inspection (See 4.4)
b. Quality Conformance Inspection (See 4.5)

4.4 First article inspection

4.4.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.4.2. The first article sample shall consist of the following items in the sample quantities indicated.

PART DESCRIPTION	DRAWING	QUANTITY
Igniter, Electric, M63, Metal Parts	9235760	25
Ignition Element Assembly	8839505	50
Igniter Assembly, Prior to Inserting Closing Plug	8839497	50
Igniter, Primer, Electric, M63	8839497	490
Sealed Barrier Bag	8837897	10

4.4.2 Inspections to be performed. See MIL-A-48078, Table I, and 4.5.2.1 through 4.5.2.12 herein.

4.4.3 Rejection. (See MIL-A-48078)

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(Copies of specifications, standards, handbooks, drawings, and

TABLE I First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-46224E (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Igniter, Primer, Electric, M63, Metal Parts	1 OF 2	9235760
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Metal Parts Examination for defects	25(a)		3.3	4.5.2.1 to 4.5.2.3

NOTES:
(a) 25 of each component.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-46224E (AR)

PARAGRAPH	TITLE: Igniter, Primer, Electric, M63		SHEET 2 OF 2		DRAWING NUMBER: See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ignition element assembly (Dwg. 8839505)				
	Examination for defects	50		3.2	4.5.2.4
	Igniter, primer, electric, M63 (prior to inserting closing plug) (Dwg 8839497)				
	Examination for defects	50		3.4	4.5.2.6
	Moisture content of charge composition	4.5.3.1.1		3.5.1	4.6.1.1
	Determination of charge composition	4.5.3.1.5		3.2	4.6.5
	Igniter, primer, electric, M63 (Dwg 8839497)				
	Examination of defects	50		3.4	4.5.2.7
	Electrical resistance check	50		3.6	4.6.2
	Functioning test	315		3.7	4.6.3
	Non-functioning test	125		3.8	4.6.4
	Sealed barrier bag (Dwg 8837897)				
	Examination for defects	10		3.2	4.5.2.10
	Heat seal test	10		3.2	4.6.6

NOTES:

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4.5 Quality conformance inspection

4.5.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of igniters shall contain:

a. Metal parts from not more than one lot interfix number from one manufacturer.
b. Ignition charge composition from not more than one lot interfix number from one manufacturer.
c. Ignition elements from not more than one lot interfix number from one manufacturer.
d. Black powder from not more than one lot from one manufacturer.

4.5.2 Examination. (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 8839503
4.5.2.1	Plug, Closing				NEXT HIGHER ASSEMBLY 9235760
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Pitch diameter of thread		0.40%	3.3	Gage
102	Hole missing or obstructed		0.40%	3.3	Visual
Minor					
201	Major diameter of thread, min.		0.65%	3.3	Gage
202	Diameter of hole, min		0.65%	3.3	Gage
203	Diameter of counterbore		0.65%	3.3	Gage
204	Total length, max.		0.65%	3.3	Gage
205	Length of shoulder, min		0.65%	3.3	Gage
206	Length of undercut, max		0.65%	3.3	Gage
207	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8839504
4.5.2.2	Body				NEXT HIGHER ASSEMBLY 9235760
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Pitch diameter of external threads		0.40%	3.3	Gage
102	Pitch diameter of internal threads		0.40%	3.3	Gage
Minor					
201	Concentricity of ignition element cavity with ignition element hole		0.65%	3.3	Gage
202	Length of thread undercut, max.		0.65%	3.3	Gage
203	Distance to external threads, max.		0.65%	3.3	Gage
204	Major diameter of external threads, min.		0.65%	3.3	Gage
205	Minor diameter of internal threads, max.		0.65%	3.3	Gage
206	Length of internal threads, min.		0.65%	3.3	Gage
207	Diameter of ignition element cavity		0.65%	3.3	Gage
208	Diameter of ignition element hole		0.65%	3.3	Gage
209	Location of wrench holes		0.65%	3.3	Gage
210	Diameter of flange		0.65%	3.3	Gage
211	Total length		0.65%	3.3	Gage
212	Depth of ignition element hole		0.65%	3.3	Gage
213	Thickness of flange		0.65%	3.3	Gage
214	Diameter of wrench hole, min.		0.65%	3.3	Gage
215	Depth of wrench holes		0.65%	3.3	Gage
216	Diameter of body, max.		0.65%	3.3	Gage
217	Evidence of poor workmanship including absence of chamfer or radius		0.65%	3.9	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8839509
4.5.2.3	Retainer				NEXT HIGHER ASSEMBLY 9235760
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Pitch diameter, min.		0.40%	3.3	Gage
102	Major diameter, min.		0.40%	3.3	Gage
103	Diameter of hole, min.		0.40%	3.3	Gage
Minor					
201	Thickness, min.		0.65%	3.3	Gage
202	Slot missing		0.65%	3.3	Visual
203	Surface finish improper		0.65%	3.3	Visual
204	Pitch diameter, max.		0.65%	3.3	Gage
205	Evidence of poor workmanship		0.65%	3.9	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.5.2.4	Ignition Element Assembly	1 OF 1	8839505
			NEXT HIGHER ASSEMBLY: 8839497

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Charge above flush		100.0%	3.2	Visual
2	Charge composition on outside surface		100.0%	3.2	Visual
Major					
101	Charge missing or insufficient		0.40%	3.2	Visual
102	Moisture content	4.5.3.1.1		3.5.1	4.6.1.1
103	Analysis of ignition element charge composition	4.5.3.1.5		3.2	4.6.5
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8839502 / NEXT HIGHER ASSEMBLY 8839497
4.5.2.5	Closing Plug Loading Assembly				
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Charge missing		0.40%	3.2	Visual
102	Moisture content of black powder charge		4.5.3.1.2	3.5.2	4.6.1.2
103	Disc or tape missing, punctured or peeling		0.40%	3.2	Visual
104	Lacquer missing from disc (when applicable)		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

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MIL-I-46224E (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.5.2.6	M63 Igniter Assembly, (Prior to Inserting Closing Plug)		8839497
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Closing disc missing or damaged to extent that explosive is exposed		0.40%	3.4	Visual
102	Retainer loose		0.40%	3.4	Manual
103	Retainer stake missing (when applicable)		0.40%	3.4	Visual
104	Retainer not fully seated		0.40%	3.4	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.5.2.7	M63 Igniter Assembly				8839497
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth to electrode		0.40%	3.4	Gage
102	Flash hole obstructed		0.40%	3.4	Visual
103	Thread damaged		0.40%	3.4	Visual
Minor					
201	Total length, max		0.65%	3.4	Gage
202	Closing plug loose		0.65%	3.4	Manual
203	Marking missing, misleading or unidentifiable		0.65%	3.4	Visual
204	Evidence of poor workmanship		0.65%	3.9	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8837897
4.5.2.8	Box, Set-up, Prior to Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Igniter missing		0.40%	3.2	Visual
102	Igniter inverted		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8837897
4.5.2.9	Box, Set-up, Sealed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents loose		0.40%	3.2	Manual
102	Box damaged		0.40%	3.2	Visual
103	Sealing strip incomplete or badly wrinkled		0.40%	3.2	Visual
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-I-46224E (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8837897
4.5.2.10	Sealed Barrier Bag				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Waterproofness of barrier bag destroyed by rupture, puncture or separation of heat seal		0.40%	3.2	Visual
102	Evidence of crushed box NOTE: Package should be opened if there is evidence of crushing		0.40%	3.2	Visual
Minor 201	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied		0.65%	3.2	Visual
202	Marking of bag misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8837898
4.5.2.11	Wood Packing Box Prior to Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect number of cartons		0.40%	3.2	Visual
102	Any packing component missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-I-46224E (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8837898
4.5.2.12	Wood Packing Box Sealed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged to the extent that the contents are exposed or likely to become exposed in subsequent handling		0.40%	3.2	Visual
102	Hardware or strapping broken or loose		0.40%	3.2	Visual
Minor					
201	Hardware or strapping incorrectly assembled		0.65%	3.3	Visual
202	Metallic seal unsealed or improperly positioned		0.65%	3.3	Visual
203	Contents loose		0.65%	3.3	Manual
204	Marking misleading or unidentifiable		0.65%	3.3	Visual
205	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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4.5.3. Testing. Inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.5.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See MIL-A-48078 Section 6, and 6.2 herein.

4.5.3.1 Moisture content.

4.5.3.1.1 Ignition element charge composition. Three samples minimum shall be selected from each drying oven after completion of the drying cycle. If the moisture content of any sample exceeds the requirement, that quantity of material represented by the sample shall either be redried until acceptable results are achieved, or rejected. The selected samples must not be dried more than any other unit which could be selected.

4.5.3.1.2 Black powder. One sample, minimum, shall be selected from each eight hours production. If the moisture content of the sample exceeds the requirement, and loading has not begun, that quantity of material represented by the sample shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all assemblies loaded with the non-conforming material shall be rejected.

4.5.3.2 Electrical resistance. This test shall be performed 100 percent. Any igniter that fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.6.2.

4.5.3.3 Functioning. Beginning with the first lot produced and continuing until three consecutive lots have complied with the requirements specified, 315 samples shall be randomly selected from each lot for test as specified in 4.6.3. If two or more failures occur, the lot shall be rejected. After three consecutive lots have complied with the above criteria, a sample of 200 igniters shall be selected from each lot for this test. If two or more igniters fail to comply with requirements, the lot shall be rejected.

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4.5.3.4 Non-functioning. Beginning with the first lot produced and continuing until three consecutive lots have complied with the specified requirement, 125 samples shall be selected from each lot and tested in accordance with 4.6.4. If any igniter functions, the lot shall be rejected. Igniters subjected to this test shall not be returned to the lot. (Destructive test) After three consecutive lots have complied with the above, 50 samples shall be selected from each lot for this test. If any igniter functions, the lot shall be rejected. Assemblies subjected to this test shall not be returned to the lot. (Destructive test)

4.5.3.5 Determination of ignition charge composition. Approximately two grams of ignition charge composition shall be selected from that quantity of composition used in loading a complete lot of igniter primers. The analysis shall be performed as specified in 4.6.5. The composition lot shall be rejected if it fails to comply with the requirements.

4.5.3.6 Heat seal test of sealed bag. Sampling, acceptance criteria and test procedures for this test shall be as specified in MIL-P-116.

4.6 Methods of inspection

4.6.1. Moisture content

4.6.1.1 Igniter charge composition. The moisture content of the ignition charge composition shall be determined as specified in Method 102.2.1 of MIL-STD-1234 except that the electric ignition element loading assembly and only a two hour drying time shall be used. The upper insulator may be removed from the electric ignition element loading assembly prior to analysis.

4.6.1.2 Black powder. The moisture content of the black powder shall be determined in accordance with the procedure specified in MIL-P-223.

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4.6.2 Electrical resistance. The igniter shall be tested for electrical resistance by applying an electrical current of twenty milliamperes, maximum, using the electrode as one contact point and the body as the other.

4.6.3 Functioning. The igniter shall be tested for electrical functioning by applying an electrical current of 1.25 amperes, maximum, using the electrode for one contact point and the body for the other.

4.6.4 Non-functioning. The igniter shall be tested for non-functioning by applying an electrical current of 190 to 200 milliamperes for one second, minimum, using the electrode as one contact point and the body as the other.

4.6.5 Determination of ignition charge composition. The chemical composition of the ignition charge shall be considered satisfactory if it is in accordance with the following:

Component	Percentage
Potassium Chlorate and Gum Solids	45.4 $\pm$ 2.1%
Lead Thiocyanate	34.9 $\pm$ 2.0%
Charcoal	19.7 $\pm$ 1.5%

4.6.5.1 Charcoal. Transfer approximately 2 grams of the igniter charge composition into a tared medium porosity crucible. Place in oven at 70° C overnight to drive off water. Place crucible in desiccator to cool and then weigh to establish sample weight. Wash with ten 25 ml portions of 5% potassium nitrate solution followed by 50 ml of distilled water. Use a small glass rod to break up any lumps in the sample during the extraction process. Preserve the filtrate for the determination for lead thiocyanate. Place in oven maintained at 100 $\pm$ 5 °C for 2 hours. Cool in a desiccator and weigh. Calculate the percent charcoal as follows:

$$\text{Percent charcoal} = \frac{100\ (C - B)}{A - B}$$

A = Sample weight before washing (including crucible)
B = Empty crucible weight
C = Sample weight after washing

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4.6.5.2 Lead thiocyanate. Exactly 35 ml of standardized 0.1 normal (N) silver nitrate shall be added to the filtrate reserved from the charcoal determination. The flask and contents shall be swirled and 100 ml of 3N nitric acid shall be added. The nitric acid used should be water white and free of oxides of nitrogen. The swirling of the mixture shall be continued for five minutes, 5 ml of a saturated solution of ferric ammonium sulfate added and the solution titrated with standardized 0.1N potassium thiocyanate solution to the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follows:

$$\text{Percent lead thiocyanate} = \frac{1.617\ (A - B)}{C}$$

A = Volume of silver nitrate solution
B = Volume of potassium thiocyanate
C = Weight of sample

4.6.5.3 Potassium chlorate and gum solids. The percentage of potassium chlorate and gum solids is obtained by subtracting the total percentage of charcoal and lead thiocyanate from one hundred percent.

4.6.6 Barrier bag heat seal test. The procedure for performing the test for security of the heat seal shall be as specified in Mil-P-116.

5 PACKAGING

5.1 Preservation and packaging

5.1.1 Igniter parts. Packaging shall be in accordance with MIL-STD-1169 Level C requirements.

5.1.2 Igniters. Preservation and packaging shall be in accordance with Dwg. 8837897, Level C requirements.

5.2 Packing

5.2.1 Igniter parts. Packing shall be in accordance with MIL-STD-1169 Level C requirements.

5.2.2 Igniters. Cartons shall be packed in accordance with Dwg. 8837898 Level C requirements.

5.3 Marking. Igniter containers and bags shall be marked in accordance with instructions contained in Dwgs. 8837897 and 8837898. Igniter parts containers shall be marked in accordance with MIL-STD-1169 requirements.

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6. NOTES

6.1 Acquisition requirements. See MIL-A-48078.

6.2 Submission of inspections equipment designs for approval. See MIL-A-48078. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, NJ, 07806-5000. This address will be specified on the Contract Data Requirements List, DD form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.3 Check test for possible deterioration. The following test is designed to prevent igniters that have deteriorated from being assembled into major ammunition items. The test shall be performed on igniters from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the igniters are assembled does not require that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Igniter assemblies shall be selected from each lot for this test.

6.3.1 Functioning. 125 igniters shall be selected from the lot and subjected to the test specified in 4.6.3. If 2 or more igniters fail to comply with the requirement, the lot shall be rejected.

6.3.2 Non-functioning. 80 igniters shall be selected from the lot and subjected to the test specified in 4.6 4. If any igniter functions the lot shall be rejected.

6.3.3 Electrical resistance. Each igniter in the lot shall be tested as specified in 4.6.2. Any igniter that fails to comply with the requirement shall be classed defective and removed from the lot.

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6.4 Subject term (key word) listing.

Ammunition
Tank
Cannon
Cartridge
Primer
Igniter

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian
Army-AR

Preparing activity
Army-AR

(Project No. 1390-A647)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-46224E (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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